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ORF307: Optimization

Fall Semester 2025 – 1st Midterm Exam

October 9, 10:40 am – 12:00 pm

Exam Rules

1. Please write out and sign the following pledge on the solution booklet: "I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam."
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. There are 3 problems worth a total of 100 points.
6. At the end of the exam, please turn in both your solution booklet and the exam questions.

Problem 1 *(30 points). Modeling Humanitarian Support under Constraints*

Following a major earthquake, a humanitarian organization must procure three types of supplies - food (x_1), water (x_2), and medical kits (x_3) - to support four affected regions: R_1, R_2, R_3 and R_4. The decision variables $x = (x_1, x_2, x_3)^T \in \mathbb{R}^3$ represents the quantities of each supply to be purchased. The organization uses a known matrix $A \in \mathbb{R}^{4\times 3}$ that maps the procurement vector x to the utility delivered to each region: $\hat{b} = Ax$, where $\hat{b} \in \mathbb{R}^4$ and $\hat{b}_i$ represents the utility received by region R_i, for $i = 1, 2, 3, 4$. The governor of each region has requested a specific utility level to be delivered to her/his region. The organization captures these requests in the demand utility vector $b \in \mathbb{R}^4$.

The organization's main goal is to match, as closely as possible, the delivered utility to the requested utility, subject to the following constraints:

- **Budget constraint**: Given that each unit of supply i costs c_i dollars, the total procurement cost must not exceed the organization's annual budget β.
- **Policy constraint (Region Priority)**: Region R_2 is identified as especially vulnerable and must receive adequate support. For this reason, the utility delivered to R_2 must be at least 90% of the utility level requested by its governor.

  Note: The utility delivered to each region is determined by a weighted combination of the supplies procured. Specifically, the utility delivered to region R_i corresponds to the inner product of the i-th row of matrix A with the procurement vector x.
- **Minimum Total Utility**: The sum of the utilities delivered to all four regions must be greater than or equal to a known critical threshold, U_{min}.
- **Nonnegativity**: Negative procurement is not allowed.

Your tasks for this problem are:

a. (10 points) Formulate a constrained optimization problem that minimizes the L_1-norm of the deviations between delivered and requested utility.

b. (5 points) Reformulate the optimization problem in Part a. as an LP problem.

c. (10 points) Formulate a constrained optimization problem that minimizes the L_∞-norm of the deviations between delivered and requested utility (the constraints are the same as in Part a.)

d. (5 points) Reformulate the optimization problem in Part c. as an LP problem.

Solution.

As described in the problem the delivered utility is $\hat{b} = Ax$, and the requested utility is $b \in \mathbb{R}^4$. We define the deviation vector $d := Ax - b$.

a Our aim is to minimize $\|Ax - b\|_1$, subject to the given constraints.

$$
\begin{aligned}
\min_{x \in \mathbb{R}^3} \quad & \|Ax - b\|_1 \\
\text{s.t.} \quad & c^T x \leq \beta \quad \text{(Budget constraint)} \\
& a_2^T x \geq 0.9 b_2 \quad \text{(Region } R_2 \text{ priority)} \\
& \mathbf{1}^T A x \geq U_{\min} \quad \text{(Minimum total utility)} \\
& x \geq 0 \quad \text{(Nonnegativity)}
\end{aligned}
$$

b. LP Reformulation:

Introduce auxiliary variables $u \in \mathbb{R}^4$ such that $u_i \geq |(Ax - b)_i|$ for all $i = 1, 2, 3, 4$.

$$
\begin{aligned}
\min_{(x,u) \in \mathbb{R}^7} \quad & \sum_{i=1}^{4} u_i \\
\text{s.t.} \quad & Ax - b \leq u \\
& -(Ax - b) \leq u \\
& c^T x \leq \beta \\
& a_2^T x \geq 0.9 b_2 \\
& \mathbf{1}^T A x \geq U_{\min} \\
& x \geq 0, \quad u \geq 0
\end{aligned}
$$

c. In this case we aim to minimize $\|Ax - b\|_\infty$, subject to the same constraints.

$$
\begin{aligned}
\min_{x \in \mathbb{R}^3} \quad & \|Ax - b\|_\infty \\
\text{s.t.} \quad & \text{Same constraints as above}
\end{aligned}
$$

d. LP Reformulation

Introduce a scalar variable $t \in \mathbb{R}$ such that $t \geq |(Ax - b)_i|$ for all i.

$$
\begin{aligned}
\min_{(x,t) \in \mathbb{R}^4} \quad & t \\
\text{s.t.} \quad & Ax - b \leq t \cdot \mathbf{1} \\
& -(Ax - b) \leq t \cdot \mathbf{1} \\
& c^T x \leq \beta \\
& a_2^T x \geq 0.9 b_2 \\
& \mathbf{1}^T A x \geq U_{\min} \\
& x \geq 0, \quad t \geq 0
\end{aligned}
$$

Problem 2 *(35 points). Scheduling with Managerial Constraints*

Alice has just started her summer internship as an Optimization Specialist in a company that manufactures two products: P1 and P2. One unit of each product needs 1 hour to be produced. The company earns \$1 and \$1.5 of profit when it sells one unit of products P1 and P2, repsectively. John, the manager of the company, gave Alice the task of optimizing the weekly production scheduling of these two products. He told her that she has up to 100 hours of machine time available every week and this time should be allocated diligently between the two products so that the total profit of the company is maximized.

John told Alice that due to some contractual obligations and safety requirements, she must allocate at least 20% of the time to product P1 and no more than 40% of the time to product type P2. Another additional requirement that John emphasized was that the time allocated to either product should not exceed the time allocated to the other by more than 30% of the total available time.

a. (10 points) Formulate a linear programming problem to maximize the total profit of the company subject to the constraints that the manager specified.

b. (10 points) Plot the feasible region defined by the constraints. Compute the coordinates of all the vertices of the feasible region.

c. (10 points) Draw three contour lines of the objective function that pass through the feasible region. Identify the optimal solution and briefly justify it graphically. Compute also the optimal objective value.

d. (5 points) After plotting the feasible region can you identify any constraints that may be redundant (a constraint is redundant of it does not affect the feasible region at all)? Briefly explain your reasoning.

Solution.

(a) Formulation of the Linear Program

Our decision variables are x_1 and x_2 and they represent the number of hours allocated to products P1 and P2, respectively. They are both non-negative, $x_1, x_2 \geq 0$. In addition, since each unit of P1 and P2 require 1 hour to produce, we can easily map hours and number of products produced. Therefore, since our aim to maximize our total profit, our objective function is defined as:

$$\max \quad x_1 + 1.5x_2$$

Based on the requirements that the manager gave to Alice, we can formulate the following constraints:

$$
\begin{array}{ll}
x_1 + x_2 \leq 100 & \text{(total production machine time)} \\
x_1 \geq 0.2 \cdot 100 = 20 & \text{(at least 20\% of time to P1)} \\
x_2 \leq 0.4 \cdot 100 = 40 & \text{(at most 40\% time to P2)} \\
x_1 - x_2 \leq 30 & \text{(time allocated to P1 cannot exceed P2 by more than 30 hours)} \\
x_2 - x_1 \leq 30 & \text{(time allocated to P2 cannot exceed P1 by more than 30 hours)} \\
x_1, x_2 \geq 0 & \text{(nonnegativity constraints)}
\end{array}
$$

(b) Plotting the Feasible Region

To plot the feasible region, we have to first plot all the half-spaces defined by the managerial constraints and then take their intersection. Figure 1 shows all the constraints and the feasible region as the gray area. It is a bounded polyhedron (polytope), defined by the following five vertices:

$$V_1 = (20, 40), V_2 = (60, 40), V_3 = (65, 35), V_4 = (30, 0), V_5 = (20, 0)$$

NOTE: As we discussed in class, to draw the half space defined by the inequality $a_1 x_+ a_2 x_2 \leq b$, first we draw the hyperplane $a_1 x_+ a_2 x_2 = b$. Next, we compute the gradient of the affine function $\nabla(a_1 x_1 + a_2 x_2 - b) = (a_1, a_2)^T$. This gradient (which is a vector) gives us the direction of maximum increase of the function $a_1 x_1 + a_2 x_2 - b$. Since our inequality is a *less-than inequality* ($\leq$) we take direction of the negative gradient (i.e., $-\nabla(a_1 x_+ a_2 x_2) = -(a_1, a_2)^T$) to determine the half-space defined by the inequality $a_1 x_+ a_2 x_2 \leq b$.

If we had a greater-than inequality (e.g., $a_1 x_+ a_2 x_2 \geq b$) we would have used the direction defined by the gradient itself to determine the half-space (i.e., $\nabla(a_1 x_1 + a_2 x_2 - b) = (a_1, a_2)^T$).

(c) Contour Lines and Optimal Solution

The contour lines of the objective function $x_1 + 1.5 x_2 = z$ are straight lines with slope $-\frac{1}{1.5}$. Also, we recall that the gradient of the objective function, $\nabla(x_1 + 1.5 x_2) = (1, 1.5)^T$, defines the direction of its maximum increase. The gradient is also perpendicular to the contours. Then to find the optimal solution graphically we follow the steps:

- Draw several contour lines of the objective function (in this problem we draw three contours) along the direction of the gradient vector.
- Slide the line in the direction of increasing profit (i.e., the gradient vector) while remaining tangent to the feasible region.
- The optimal solution will be at the vertex of the feasible region where the contour line of the objective is **tangent for the last time before it separates from the feasible region**.

Figure 1 shows the contours of the objective for three different values of $z = 60, 90, 120$. The optimal solution is the vertex, V_2, with coordinates $(x_1^*, x_2^*) = (60, 40)$, and the optimal objective value is $z^* = x_1^* + 1.5x_2^* = 120$.

(d) Identifying the Redundant Constraint After plotting the feasible region we can observe that the constraints $x_1 \geq 0$ **and** $x_2 - x_1 \leq 30$ do not affect the formation of the feasible region. The constraints that make them redundant are: $x_1 \geq 20$ and $x_2 \leq 40$.

Alice can highlight this to her manager to streamline the model and reduce unnecessary constraints.

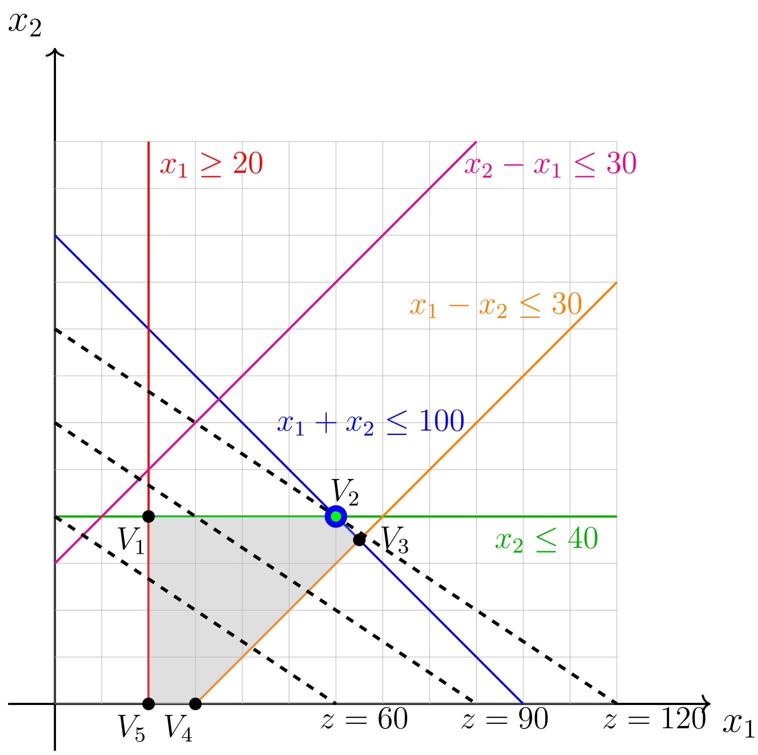

Figure 1: The feasible region is the areas shaded in gray. Vertices are marked with large black dots, with the optimal point $(60, 40)$ highlighted in blue and green to emphasize that it is at the intersection of blue and the green hyperplanes. The dashed lines show contour levels of the objective function $z = x_1 + 1.5x_2$, for $z = 60, 90, 120$.

Problem 3 *(35 points). Regularized Weighted Least Squares.*

Consider the least squares optimization problem

$$\min_{x\in\mathbb{R}^n} \quad \mu\|Ax-b\|^2 + \lambda\|x\|^2 + \|Cx-d\|^2 \tag{1}$$

where $A \in \mathbb{R}^{m\times n}, b \in \mathbb{R}^m, C \in \mathbb{R}^{p\times n}, d \in \mathbb{R}^p$.

a. *(10 points)* Define a new matrix $\tilde{A}$ and a new vector $\tilde{b}$ such that the optimization problem (1) can be written in the standard least squares form:

$$\min_{x\in\mathbb{R}^n} \quad \|\tilde{A}x-\tilde{b}\|^2 \tag{2}$$

Also, state the exact dimensions of the matrix $\tilde{A}$ and vector $\tilde{b}$.

b. *(10 points)* Derive the normal equations for the least squares problem in standard form defined by (2) and next express them in terms of the matrices and vectors (A, b, C, d) and parameters μ, λ of the original least squares problem defeined by (1).

c. *(15 points)* Now suppose we add the equality constraints $Fx = h$ to define the constrained least squares problem:

$$\begin{aligned} \min_{x\in\mathbb{R}^n} \quad & \|\tilde{A}x-\tilde{b}\|^2 \\ s.t. \quad & Fx = h \end{aligned} \tag{3}$$

where $F \in \mathbb{R}^{q\times n}$ and $h \in \mathbb{R}^q$. Define the Lagrangian function of this problem and write the KKT conditions. From the KKT conditions derive the optimal solution $\tilde{x}^*$ of problem (3).

Note: All norms in the problem are L_2 (Euclidean) norms.

Solution.

a. Let us define the objective function as

$$f(x) = \mu\|Ax-b\|^2 + \lambda\|x\|^2 + \|Cx-d\|^2 \tag{4}$$

We note that the objective consists of three terms all described by L_2-norms squared. We can write each term as follows:

$$\begin{aligned} f_1(x) &= \mu\|Ax-b\|^2 = \|\sqrt{\mu}(Ax-b)\|^2 \\ f_2(x) &= \lambda\|x\|^2 = \|\sqrt{\lambda}x\|^2 \\ f_3(x) &= \|Cx-d\|^2 \end{aligned} \tag{5}$$

Then we can define the matrix $\tilde{A}$ and the vector $\tilde{b}$ as follows

$$\tilde{A} = \begin{bmatrix} \sqrt{\mu}A \\ \sqrt{\lambda}I_n \\ C \end{bmatrix}, \qquad \tilde{b} = \begin{pmatrix} \sqrt{\mu}b \\ 0 \\ d \end{pmatrix}$$

We can now show that $\|\tilde{A}x - \tilde{b}\|^2$ equals the objective $f(x)$ by expanding the norm and performing the algebra

$$\|\tilde{A}x - \tilde{b}\|^2 = (\tilde{A}x - \tilde{b})^T(\tilde{A}x - \tilde{b}) = x^T\tilde{A}^T\tilde{A}x - 2\tilde{b}^T\tilde{A}x + \tilde{b}^T\tilde{b} \tag{6}$$

Computing the matrix/matrix and matrix/vector products we get

$$x^T\tilde{A}^T\tilde{A}x = x^T \begin{bmatrix} \sqrt{\mu}A^T & \sqrt{\lambda}I_n & C^T \end{bmatrix} \begin{bmatrix} \sqrt{\mu}A \\ \sqrt{\lambda}I_n \\ C \end{bmatrix} x = \mu x^T A^T A x + \lambda x^T x + x^T C^T C x$$

$$2\tilde{b}^T\tilde{A}x = 2\begin{pmatrix} \sqrt{\mu}b^T & 0^T & d^T \end{pmatrix} \begin{bmatrix} \sqrt{\mu}A \\ \sqrt{\lambda}I_n \\ C \end{bmatrix} x = 2\mu b^T A x + 2d^T C x$$

$$\tilde{b}^T\tilde{b} = \begin{pmatrix} \sqrt{\mu}b^T & 0^T & d^T \end{pmatrix} \begin{pmatrix} \sqrt{\mu}b \\ 0 \\ d \end{pmatrix} = \mu b^T b + d^T d$$

Adding the last three equations together we obtain the objective function $f(x)$.

The dimensions of the matrix $\tilde{A}$ are $(m+n+q) \times n$, and the dimension of the vector $\tilde{b}$ is $(m+n+q)$. In mathematical notation we have: $\tilde{A} \in \mathbb{R}^{(m+n+q)\times n}$ and $\tilde{b} \in \mathbb{R}^{(m+n+q)}$.

b. We can compute the optimal solution of the unconstrained least squares problem

$$\min_x \ \|\tilde{A}x - \tilde{b}\|^2 \tag{7}$$

by setting the gradient of the objective function to zero and deriving the normal equations:

$$\nabla(\|\tilde{A}x - \tilde{b}\|^2) = 0 \implies \tilde{A}^T\tilde{A}x = \tilde{A}^T\tilde{b}$$

From part (a) we have computed the terms $\tilde{A}^T\tilde{A}$ and $\tilde{A}^T\tilde{b}$, which can substitute to the normal equations:

$$(A^TA + \lambda I_n + C^TC)x^* = \mu A^T b + C^T d$$

Solving the normal equations we get the optimal solution of problem (1), provided that $(A^TA + \lambda I_n + C^TC)$ is invertible:

$$x^* = (A^TA + \lambda I_n + C^TC)^{-1}(\mu A^T b + C^T d)$$

However, in order to invert we must make sure that the coefficient matrix is invertible. The matrix $(A^TA+\lambda I_n+C^TC)$ is the sum of two positive semidefinite definite matrices (A^TA and C^TC) and a positive definite matrix (λI_n), and therefore it is itself positive definite and therefore invertible. We can hence conclude that the optimal solution of the least squares problem is

c. We define the Lagrangian function of the constrained least squares problem:

$$L(x,z) = \|\tilde{A}x - \tilde{b}\|^2 + z^T(Fx - h)$$

where z is the vector of the Lagrange multipliers. The KKT conditions are obtained by computing the gradient of the Lagrangian L and setting it to zero:

$$\nabla L(x,z) = \begin{bmatrix} \nabla_x L(x,z) \\ \nabla_z L(x,z) \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Let's compute each element of $\nabla L(x,z)$ separately.

$$\nabla_x L(x,z) = 2\tilde{A}^T\tilde{A}x^* - 2\tilde{A}^T\tilde{b} + F^T z^* = 0 \implies 2\tilde{A}^T\tilde{A}x^* + F^T z^* = 2\tilde{A}^T\tilde{b} \tag{8}$$

$$\nabla_z L(x,z) = Fx^* - h = 0 \implies Fx^* = h \tag{9}$$

OPTION 1:

We can find the optimal solution x^* by solving system of linear equations (8) and (9) as a block system of linear equations that involves x^* and z^* together. In particular, the block system has the following form:

$$\begin{bmatrix} 2\tilde{A}^T\tilde{A} & F^T \\ F & 0 \end{bmatrix} \begin{pmatrix} x^* \\ z^* \end{pmatrix} = \begin{pmatrix} 2\tilde{A}^T\tilde{b} \\ h \end{pmatrix} \tag{10}$$

We know from the lectures that the coefficient matrix

$$\begin{bmatrix} 2\tilde{A}^T\tilde{A} & F^T \\ F & 0 \end{bmatrix}$$

is invertible if the matrix F has linear independent rows and the block matrix

$$\begin{bmatrix} A \\ F \end{bmatrix}$$

has linear independent columns. So in order to invert the coefficient matrix we assume that these conditions are satisfied. Hence the solution vector x^* is given by the first n elements of the vector

$$\begin{pmatrix} x^* \\ z^* \end{pmatrix} = \begin{bmatrix} 2\tilde{A}^T\tilde{A} & F^T \\ F & 0 \end{bmatrix}^{-1} \begin{pmatrix} 2\tilde{A}^T\tilde{b} \\ h \end{pmatrix}$$

OPTION 2:

We can solve the system of linear equations (8) and (9) explicitly and can directly compute x^* (the same assumptions as in OPTION 1 must also hold). We first solve (8) with respect to x^*

$$x^* = (\tilde{A}^T \tilde{A})^{-1} \tilde{A}^T \tilde{b} - \frac{1}{2} (\tilde{A}^T \tilde{A})^{-1} F^T z^* \tag{11}$$

This formula for x^* still contains the unknown z^*. Substituting (11) to (9) we can compute z^*

$$z^* = 2(F(\tilde{A}^T \tilde{A})^{-1} F^T)^{-1} (F(\tilde{A}^T \tilde{A})^{-1} \tilde{A}^T \tilde{b} - h) \tag{12}$$

Substituting (12) to (11), and performing the algebraic calculations, we obtain the clean formula for x^*

$$x^* = (\tilde{A}^T \tilde{A})^{-1} \tilde{A}^T \tilde{b} - (\tilde{A}^T \tilde{A})^{-1} F^T (F(\tilde{A}^T \tilde{A})^{-1} F^T)^{-1} (F(\tilde{A}^T \tilde{A})^{-1} \tilde{A}^T \tilde{b} - h) \tag{13}$$